

Perishable Lot-Sizing with Period-Dependent Shelf Lives

Compact Mixed-Integer Programming Model (Freshest-First)

Indices and Sets

- $T = \{1, 2, \dots, \mathbf{T}\}$: set of planning periods.
- $I = \{1, 2, \dots, \mathbf{I}\}$: set of items.
- $t \in T$: production period, $u \in T$: consumption period.

Perishability

Shelf life. Units of i produced at t can be used at $u \geq t$ only if

$$u - t \leq m_{it}.$$

Arc set. Variables x_{itu} exist only when $t \leq u \leq t + m_{it}$.

Parameters

d_{iu}	demand of item i in period u .
c_{it}	unit production cost of item i in period t .
h_{it}	unit holding cost of item i per period of age.
s_{it}	setup cost of item i in period t .
κ_t	total production capacity in period t .
p_{it}	(we said only cap but if we want we can keep cap across items constant anyway so I kept this notation) per-item capacity in period t .
m_{it}	shelf life (periods) for production (i, t) .
W	(if given) warehouse capacity (end-of-period inventory).
μ_{it}	tight big- M constant for item i in period t , equal to the total demand within its consumption window: $\mu_{it} = \sum_{u-t \leq m_{it}} d_{iu}$.
g_{itu}	generalized unit cost for item i produced in t and to consuming it in period u , defined as $g_{itu} = c_{it} + h_{it}(u - t)$.

Decision Variables

$X_{itu} \geq 0$	quantity of i produced at t and consumed at u (only if $t \leq u, t + m_{it} - 1$).
$Y_{it} \in \{0, 1\}$	setup decision for item i in period t .
$S_{iu\tau} \geq 0$	freshest-first slack for item i , demand period u , threshold τ

Objective

$$\min \sum_{i=1}^{\mathbf{I}} \sum_{t=1}^{\mathbf{T}} \sum_{t \leq u \leq t+m_{it}} g_{itu} X_{itu} + \sum_{i=1}^{\mathbf{I}} \sum_{t=1}^{\mathbf{T}} s_{it} Y_{it} \quad (1)$$

Constraints

Global capacity (per production period). For all $t = 1, \dots, \mathbf{T}$,

$$\sum_{i=1}^{\mathbf{I}} \sum_{t \leq u \leq t+m_{it}} X_{itu} \leq \kappa_t. \quad (2a)$$

Capacity across items is constant (only time matters). For all $i = 1, \dots, I$, $t = 1, \dots, T$,

$$\sum_{i=1}^I X_{it} \leq p_t$$

Setup linking (big- M tied to (i, t)). For all it ,

$$\sum_{t \leq u \leq t+m_{it}} X_{itu} \leq \mu_{it} Y_{it} \quad (2b)$$

Warehouse capacity (if given). For all $u = 1, \dots, \mathbf{T} - 1$,

$$\sum_{i=1}^{\mathbf{I}} \sum_{t=1}^u \sum_w X_{itw} \leq W \quad (2c)$$

Demand balance (no backorders). For all $i = 1, \dots, \mathbf{I}$, $u = 1, \dots, \mathbf{T}$

$$\sum_{\substack{t=1, \dots, u \\ t \leq u \leq t+m_{it}}} X_{itu} = d_{iu} \quad (2d)$$

Freshest-first (LEFO). For all $i = 1, \dots, \mathbf{I}$, $u = 1, \dots, \mathbf{T}$, and thresholds $\tau = 1, \dots, u - 1$:

$$S_{iu\tau} \geq d_{iu} - \sum_{\substack{t=\tau+1, \dots, u \\ t \leq u \leq t+m_{it}}} X_{itu} \quad (2e)$$

$$(2f)$$

$$\sum_{\substack{t=1, \dots, \tau \\ t \leq u \leq t+m_{it}}} X_{itu} \leq S_{iu\tau} \quad (2g)$$

Meaning: Older cohorts ($t \leq \tau$) are used only to cover the residual after allocating from newer cohorts ($t \geq \tau+1$) towards d_{iu} .

Domains.

$$X_{itu} \geq 0, \quad Y_{it} \in \{0, 1\}, \quad S_{iu\tau} \geq 0, \quad (2h)$$